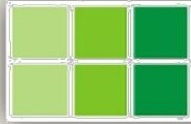




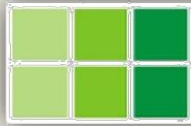
GOA ELECTRONICS LIMITED

A Subsidiary of EDC Ltd. (A Government of Goa Undertaking)

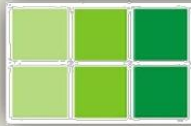
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**REVISION HISTORY**

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	01-July-2024	Gayetri Tirodkar (SSD)	Anant Yende (CTO)	New SOP

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INTRODUCTION

Database management involves a wide range of activities focusing on ensuring optimized performance, database security and availability.

1. Data Storage

- 1.1 The database files cannot be stored on system drive. (i.e. C drive).
- 1.2 Database backup cannot be stored on system drive or external data drive.
- 1.3 If the database is going to grow exponentially ensure creating separate tablespace for tables in different drives instead of using only one.

2 Access Control

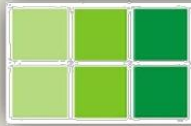
- 2.1 Implement role-based access control (RBAC) to ensure only authorized users can access or modify data.
Use the principle of least privilege.
- 2.2 Separate users to be made with limited privileges and with strong passwords.
 - a) app user :- The app user need to be separate for each applications.
 - b) view user :- There should be separate users to access the available views.
 - c) admin user:- Admin user will all admin privileges need to create with complex password.

3. Data Privacy/Security

- 3.1 Encrypt sensitive data both at rest and in transit to protect against unauthorized access. Use AES encryption standards.
- 3.2 Ensure that store procedures , functions , views and triggers do not expose sensitive data like Aadhar no , PAN No, EPIC No, Passport No, Account No, Mobile No, Telephone No, Email Id.

4. Audit Logs

- 4.1 Enable Auditing and logging to monitor access and changes to data. Regularly review logs for suspicious activities.
- 4.2 Maintain the exception/error logs for reviewing.
- 4.3 Maintain access logs to check if any query taking more time for execution.



5. Performance Maintaining and Optimization

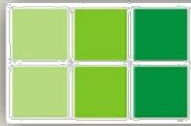
- 5.1 Create indexes on columns that are frequently searched or used in joins to improve query performance. Be mindful of index overhead on write operations.
- 5.2 Clear log files after one month.
- 5.3 Monitor transactions which take longer than 2 minutes on weekly basis and take up for optimization.
- 5.4 Find out missing index and ensure only required indexes are added. Any unused indexes to be removed.
- 5.5 Check tables without primary key and get the same added/reviewed.
- 5.6 On high CPU usage check active connections and check which processes can be terminated. Also check if any job doesn't close in due time and running. Also check for deadlocks if any.
- 5.7 Check with TL which tables can be archived which is heavier than 1 GB.
- 5.8 Check tables which can be skipped from daily backup which are heavier than 10GB and of which data doesn't change over time.

6. Backup and Recovery

- 6.1 Regular Backups: The db backup should be compulsorily scheduled as per below mentioned frequency:

Sr. No.	Backup Time	Backup Schedule
1	Less than 5 Seconds	Every 30 Minutes
2	Less than 30 Seconds	Every 60 Minutes
3	From 30 Seconds to 5 Minutes	Every 6 Hours (1am ,7am ,1pm , 7pm)
4	More than 5 Minutes and less than 60 Minutes	Every night 1am
5	More than 120 Minutes	Once in a week full backup and hourly incremental backup

- 6.2 Re-indexing of database need to be done on weekly basis at 12am or as per CTO approval.
- 6.3 The dead tuples/ defragmentation to be carried out daily at 11pm.
- 6.4 Develop and test a disaster recovery plan to ensure quick recovery in case of catastrophic failures.



7. Scalability and Maintenance

- 7.1 Capacity Planning: Monitor database usage and plan for capacity expansion to handle growth. Consider both vertical (upgrading hardware) and horizontal (adding more nodes) scaling strategies.
- 7.2 Regular Maintenance: Perform routine maintenance tasks such as database vacuuming, index rebuilding, and update statistics to ensure optimal performance.

8. Compliance and Documentation

- 8.1 Documentation: Maintain comprehensive documentation of database schema, relationships, and configuration settings. Update documentation following any changes to the database system or standards and maintain version control.

9. Development Practices

- 9.1 Development Environments: Use separate development, testing, and production environments to ensure stability and minimize the risk of introducing issues to production.
- 9.2 Version Control: Use version control for database schema changes and scripts to track modifications and collaborate effectively with team members.

10. Monitoring and Troubleshooting

- 10.1 Performance Monitoring: Implement tools to monitor database performance metrics (e.g. CPU usage, query performance). Set up alerts for unusual activity or performance degradation.
- 10.2 Troubleshooting: Develop and follow procedures for diagnosing and resolving database issues. Maintain a knowledge base of common problems and solutions.

11. Automation

- 11.1 Automate Repetitive Tasks: Use automation tools for routine tasks like backups, monitoring, and report generation to reduce manual effort and human error.
- 11.2 Configuration Management: Employ configuration management tools to manage and deploy database configurations consistently across environments.

Also Read

Database Coding Standards & Best Practices.